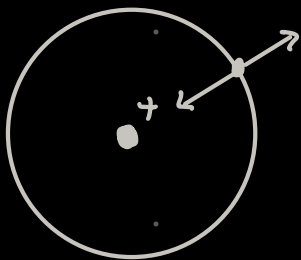


Bohr postulada: $n\hbar = mvr$



Descobrimos $a_0, r_{n,z}$



Do postulada $\frac{n\hbar}{mr} = v$ (II)

$$\frac{kze^2}{r^2} = \frac{mv^2}{r}$$

$kze^2 = mv^2 r$ (I) Substituindo (II) $\rightarrow$ (I)

$$kze^2 = m \left(\frac{n^2 \hbar^2}{m^2 r^2} \right) r \Rightarrow kze^2 = \frac{n^2 \hbar^2}{mr} \Rightarrow r = \frac{n^2 \hbar^2}{m k z e^2}$$

O raio do Bohr é $a_0 \Rightarrow n=1, z=1$

$$a_0 = \frac{\hbar^2}{m k e^2} \Rightarrow \frac{\hbar^2 4 \pi \epsilon_0}{m e^2} = 0,53 \text{ \AA}$$

$$\text{alim } r_{n,z} = a_0 \frac{n^2}{z} \approx 0,53 \text{ \AA} \frac{n^2}{z}$$

Descobrimos E_t da Orbita

$$\hat{H}_{n,z} = \hat{T}_{n,z} + \hat{V}_{n,z}$$

$$\hat{T}_{n,z} = \frac{mv_{n,z}^2}{2} ; \hat{V}_{n,z} = -\frac{kze^2}{r_0 \cdot \frac{n^2}{z}}$$

de relação de Forças em torno q

$$mv_{n,z}^2 r_{n,z} = kze^2 \Rightarrow mv_{n,z}^2 = \frac{kze^2}{r_{n,z}}$$

Valendo ao $\hat{H}$

$$\hat{H} = \frac{kze^2}{2r_0 \frac{n^2}{Z}} - \frac{kze^2}{r_0 \frac{n^2}{Z}} \Rightarrow \hat{H} = - \frac{z^2 ke^2}{n^2 r_0} = \underline{-13,6 \frac{z^2}{n^2} \text{ eV}}$$

o Comprimento de DeBroglie

einstein: $E = mc^2$

ondulatória $\left\{ \begin{array}{l} E = hf \\ c = \lambda f \end{array} \right.$

$$hf = mc^2$$

$$f = \frac{c}{\lambda}$$

$$\frac{hc}{\lambda} = mc^2 \Rightarrow \frac{h}{\lambda} = mc \text{ I momento } (\vec{p})$$

$$\lambda = \frac{h}{p} \text{ ou } \lambda = \frac{h}{mc}$$

Porc o átomo do Hidrogênio

Postulados $mvr = n\hbar$

$$mcr = n\hbar \Rightarrow mcr = n \frac{h}{2\pi}$$

$$2\pi r = n \frac{h}{mc} \Rightarrow 2\pi r = n\lambda$$

$$2\pi a_0 \frac{n}{Z} = \lambda$$

Xurupira

T.I.S.E.

$$\hat{H}\psi = \varepsilon\psi \Rightarrow -\frac{\hbar^2}{2m} \nabla^2 \psi - \frac{kZe^2}{r} \psi = \varepsilon\psi$$

$$\psi(\theta, \varphi, r) = R(r) T(\theta) F(\varphi)$$

$$-\frac{\hbar^2}{2m} \left[\overset{RTF}{\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right)} + \frac{1}{r^2 \sin^2 \theta} \overset{RTF}{\frac{\partial^2 \psi}{\partial \theta^2}} + \frac{1}{r^2 \sin \theta} \overset{RTF}{\frac{\partial}{\partial \varphi} \left(\sin \theta \frac{\partial \psi}{\partial \varphi} \right)} \right] - \frac{kZe^2}{r} \psi = \varepsilon\psi$$

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(TF r^2 \frac{dR}{dr} \right) + \frac{RF}{r^2 \sin^2 \theta} \frac{d^2 T}{d\theta^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \varphi} \left(RT \sin \theta \frac{dF}{d\varphi} \right) \right]$$

$$\underbrace{\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right)}_{\text{Ⓜ}} = r^2 \frac{d^2 R}{dr^2} + \frac{dR}{dr} \cdot 2r$$

$$-\frac{\hbar^2}{2m} \left[\frac{TF}{r^2} \left(r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) + \frac{RF}{r^2 \sin^2 \theta} \frac{d^2 T}{d\theta^2} + \frac{RT}{r^2 \sin \theta} \left(\sin \theta \frac{d^2 F}{d\varphi^2} - \frac{dF}{d\varphi} \cos \theta \right) \right]$$

$$-\frac{kZe^2}{r} R T F = \varepsilon\psi$$

$$\text{Ⓜ} \frac{TF}{r^2} \left(r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) + \text{Ⓜ} \frac{RF}{r^2 \sin^2 \theta} \frac{d^2 T}{d\theta^2} + \text{Ⓜ} \frac{RT}{r^2 \sin \theta} \left(\sin \theta \frac{d^2 F}{d\varphi^2} - \cos \theta \frac{dF}{d\varphi} \right) - \frac{kZe^2}{r} R T F = \varepsilon R T F$$

$$\frac{1}{r^2} \left(r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) - \frac{kze^2}{r} R T F - \frac{1}{r^2 \sin^2 \varphi} \frac{d^2 T}{d\varphi^2} - \frac{1}{r^2 \sin \varphi} \left(\sin \varphi \frac{d^2 F}{d\varphi^2} - \cos \varphi \frac{dF}{d\varphi} \right)$$

$$T F \left(\frac{1}{r^2} \left(r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) - \frac{R kze^2}{r} - \frac{1}{r^2} \left[\frac{F}{\sin^2 \varphi} \frac{d^2 T}{d\varphi^2} - \frac{T}{\sin \varphi} \left(\sin \varphi \frac{d^2 F}{d\varphi^2} - \cos \varphi \frac{dF}{d\varphi} \right) \right] \right)$$

$$-\frac{r^2}{A R} \left[\frac{1}{r^2} \left(r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) - \frac{R kze^2}{r A} - \frac{\epsilon R^2}{A} \right] = \frac{1}{T F} \left[\frac{F}{\sin^2 \varphi} \frac{d^2 T}{d\varphi^2} - \frac{T}{\sin \varphi} \left(\sin \varphi \frac{d^2 F}{d\varphi^2} - \cos \varphi \frac{dF}{d\varphi} \right) \right]$$

o lado esquerdo só depende de r e como o direito independe de r em a expressão é cte = C , o mesmo vale para o outro lado

Parte radial

$$-\frac{1}{R} \left(r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) + \frac{r kze^2}{A} + \frac{\epsilon R^2}{A} = C$$

Parte angular

$$\frac{1}{T \sin^2 \varphi} \frac{d^2 T}{d\varphi^2} - \frac{1}{F \sin \varphi} \left(\sin \varphi \frac{d^2 F}{d\varphi^2} - \cos \varphi \frac{dF}{d\varphi} \right) = C$$

$$\frac{1}{T \sin^2 \varphi} \frac{d^2 T}{d\varphi^2} = C + \frac{d^2 F}{F d\varphi^2} - \frac{\cos \varphi}{F} \frac{dF}{d\varphi}$$

$$\frac{d^2 T}{T d\varphi^2} = C \sin^2 \varphi + \frac{\sin^2 \varphi}{F} \frac{d^2 F}{d\varphi^2} - \sin \varphi \frac{\cos \varphi}{F} \frac{dF}{d\varphi}$$

Parte $T(\theta)$

$$\frac{d^2 T}{T d\theta^2} = -m^2$$

Parte $F(\varphi)$

$$C \sin^2 \varphi + \frac{\sin^2 \varphi}{F} \frac{d^2 F}{d\varphi^2} - \sin \varphi \frac{\cos \varphi}{F} \frac{dF}{d\varphi} = -m^2$$

Da pra brincar e isolar o nu

De novo lado esquerdo independente da direita

= $-m^2$ ~ a internet mandou chamar de m^2

$$\frac{d^2 T}{d\theta^2} = -m^2 T \quad T = A e^{im\theta} \Rightarrow \int_0^{2\pi} A e^{im\theta} \cdot A e^{-im\theta} d\theta = 1$$

$$T = \frac{1}{\sqrt{2\pi}} e^{im\theta} \quad A^2 \theta \Big|_0^{2\pi} = 1 \quad A^2 = \frac{1}{2\pi}$$

$$A = \frac{1}{\sqrt{2\pi}}$$

Desisto de resolver o resto

Só falta $F(\varphi)$ e $R(r)$

Berg Sua Missão é resolver
Para $F(\varphi)$

$$C \sin^2 \varphi + \frac{\sin \varphi}{F} \left(\sin \varphi \frac{d^2 F}{d\varphi^2} - \cos \varphi \frac{dF}{d\varphi} \right) = -m^2$$

e

Para $R(r)$

$$-\frac{1}{R} \left(r^2 + \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) + \frac{r k z e^2}{\omega} + \frac{\epsilon r^2}{\omega} = C$$

$$\frac{r k z e^2}{\omega} + \frac{\epsilon r^2}{\omega} - \frac{1}{R} \left(r^2 + \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right) = C$$

Se tudo der certo a gente vai ter 3
números Quantizados.